

HFO-1234yf

SENIOR DEISGN MAIN PROJECT REPORT

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Introduction

Problem Definition & Significance

The automotive industry is undergoing a transition away from high global warming potential (GWP) refrigerants which is being driven by regulatory mandates and increasing environmental concerns. The most common traditional auto refrigerant is currently R-134a, which has been the industry standard for mobile vehicle air conditioning (MVAC) systems for decades. R-134a has a GWP of 1,430, which makes it a significant contributor to greenhouse gas emissions (U.S. EPA, 2024). In response, international regulatory frameworks and technical assessments have emphasized the need to adopt low-GWP alternatives to mitigate climate impacts in the transportation section (U.S. EPA, 2016). Automotive companies also need a solution that not only meets these regulatory standards but also meets their own performance standards.

Hydrofluoroolefin (HFO)-1234yf has recently emerged as the leading replacement due to its extremely low GWP (<1 to 4) and short atmospheric lifetime of ~11 days, compared to its predecessor, R-134a, which has an atmospheric lifetime of 14 years (Fluorocarbons Working Group, n.d.-b; ClimatePartner, n.d.). Studies of atmospheric degradation products further support the environmental acceptability relative to other refrigerants (Fluorocarbons Working Group, n.d.-a). Therefore, R-1234yf is considered climate-safe and has a performance sufficiently comparable to R-134a (Cho et al., 2015). As a result, the adoption of R-1234yf has expanded worldwide, with virtually all new vehicles produced for the U.S. market now using this refrigerant (U.S. EPA, n.d.).

Despite the rapid shift in demand for R-1234yf, the North American production remains limited, with only two major domestic manufacturers, Honeywell and Chemours, supplying the market. With U.S. new vehicle sales predicted to exceed 16 million by the end of 2025 and each vehicle requiring 0.92 lb (0.414 kg) of R-1234yf, supply constraints are expected to intensify as adoption expands (U.S. EPA, 2016). This demand opens opportunities for new manufacturers to supply it. In order to meet the growing need and reduce reliance on geographically concentrated suppliers, additional production capacity is both economically and strategically significant for automotive original equipment manufacturers (OEMs) and the broader refrigerant supply chain.

The objective of this project is to design a competitive, environmentally responsible process for producing R-1234yf that meets the increasing OEM demand while keeping the climate impact low. By leveraging low-GWP pathways and evaluating raw material, market,

and supply chain constraints, this design aims to contribute to a more resilient and sustainable refrigerant manufacturing infrastructure.

Market Analysis

The ongoing transition from high-GWP R-134a to a low-GWP alternative has created a rapid demand for R-1234yf within the automotive sector. With U.S. new vehicle sales predicted to be around 16.1 million units by the end of 2025, it can be estimated that 80% of those vehicles will be assembled in North America. This makes the relevant market base for OEM refrigerant to be approximately 12.9 million vehicles per year (U.S. EPA, 2016). Given that R-1234yf requires roughly 8% less mass than R-134a, each vehicle would typically require 0.92 lb (0.414 kg) of refrigerant for factory fill. This corresponds to a total NA OEM demand of 11.7 million lb/year (5.3 million kg/year).

This demand is further enforced by regulatory and environmental considerations. R-1234yf has a GWP of anywhere between <1 and 4 and an atmospheric lifetime of about 11 days, which is drastically lower than the 1,430 GWP and 14 year atmospheric lifetime of R-134a (U.S. EPA, 2024; Fluorocarbons Working Group, n.d.). As a result, implementation rates have risen sharply and nearly all new vehicles in the U.S. rely on R-1234yf systems (U.S. EPA, n.d.). The shift in demand has placed a significant pressure on refrigerant manufacturers in North America to expand production capacity.

The North American market for R-1234yf is supplied by two major production facilities: Honeywell in Louisiana and Chemours in Texas. These plants represent the majority of domestic capacity and supply virtually all OEMs in the region. Both manufacturers have recently undergone massive expansions to meet the rising global and domestic demand, however, overall supply remains geographically concentrated and limited relative to projected vehicle production growth, leaving room in the market for a new supplier.

Given the total OEM demand of 5.3 million kg/year, this new manufacturing facility would target 25% of the market and would need to supply approximately 1.33 million kg/year (2.93 million lb/year) of R-1234yf. This level of production represents a significant but feasible fraction of the North American market and would enhance supply security while reducing dependence on existing manufacturers. With expected growth in global refrigerant markets and increasing regulatory emphasis on low-GWP alternatives, expanding North American production capacity supports both environmental targets and industrial competitiveness.

This supply-demand landscapes shows a clear market opportunity. A strategically located R-1234yf production facility can capture meaningful OEM market share, contribute to domestic supply resilience, and address the rising demand driven by regulatory transitions and environmental performance requirements.

Global Share

The worldwide shift from high-GWP refrigerants to low-GWP alternatives, including R-1234yf occurs because of rising environmental standards and worldwide climate change treaties. The United States Environmental Protection Agency (EPA) requires all high-GWP refrigerants to disappear from use according to their regulations while the European Union F-Gas Regulations establish equivalent requirements. The implementation of environmentally responsible refrigerants in new vehicle platforms becomes necessary because global automotive climate agreements have been established. The three largest markets for R-1234yf exist in Europe and North America and the Asia-Pacific region because these areas produce most of the world's automobiles while maintaining strict environmental regulations. The fast-paced global adoption of R-1234yf, together with rising vehicle manufacturing and expanding refrigerant substitution industry, creates an extended business potential.

Value Chain

The production value chain of R-1234yf starts with the procurement of essential raw materials which include carbon-based feedstocks and hydrogen and hydrogen fluoride (HF) and zinc and methanol. The production process transforms these materials into CFC-13 and chlorotrifluoroethylene (CTFE) and methanol and methyl chloride and the essential intermediate compound 1233xf. The manufacturing process reaches its conclusion through R-1234yf production from 1233xf followed by high-purity separation and product packaging. The finished refrigerant distribution process uses tank trucks and rail transport and pressurized storage systems which provide safe handling and delivery methods. R-1234yf finds its main application with automotive original equipment manufacturers (OEMs) and vehicle assembly plants and automotive air-conditioning system suppliers who need reliable high-purity refrigerant supplies.

Raw Material Cost

Raw Material	Price (\$ per lb)	Usage (lb per lb of 1234yf)	Cost Contribution (\$ per lb)
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C ₂ Cl ₆	0.75	2.076	1.56
HF	1.13	0.7019	0.79
Methanol	9.50	0.281	2.67
CO ₂	1.40	0.386	0.54
H ₂	2.20	0.0526	0.12
Zn	1.30	0.573	0.75

The production cost of R-1234yf amounts to \$6.43 per pound of finished product. The production cost results from using several essential raw materials which include hexachloroethane (C₂Cl₆) and hydrogen fluoride and methanol and carbon dioxide and hydrogen and zinc. The production costs of these inputs show methanol as the most expensive component because it needs to be used in large quantities at a high price point. The total expenses of the process stem from hydrogen fluoride and zinc expenses which make up most of the costs while hydrogen and carbon dioxide expenses represent the remaining portion. The complete cost analysis shows that R-1234yf production uses standard bulk chemicals which keep the material expenses of the product below its market value.

Profitability

The plant will generate 1.3 million kilograms of R-1234yf annually, which translates to 2.87 million pounds of product each year. The estimated annual raw material expenditure is \$18.46 million, based on a \$ 6.43-per-pound cost. The facility would achieve \$114.8 million in yearly sales based on a safe market value of \$40 per pound for R-1234yf. The estimated annual gross profit would be \$93.6 million after subtracting raw material costs.

Process Description

The production of R-1234yf requires two reactions. The first reaction happens when methyl chloride and CTFE are reacted with each other to form an intermediate product called 1233xf. The intermediate product is then reacted with hydrogen fluoride gas to produce our main product, R-1234yf. Since the second reaction can only be realistically done in one way, our chosen process and competing processes will instead focus on the first reaction.

Competing Process

There are four competing processes: two focus on the production of CTFE, and the other two on the synthesis of methyl chloride. For our first competing process, hexachloroethane (C_2Cl_6) is first reacted with hydrogen fluoride gas to form CFC-113 and HCl, and a slurry of zinc and methanol is then added into the mixture to form CTFE. This process is the most common method in producing CTFE, which helps us to obtain the necessary information for simulation and further research; however, this process produces zinc chloride sludge as a by-product, which is heavy and extremely corrosive, causing the waste treatment cost to be extremely high.

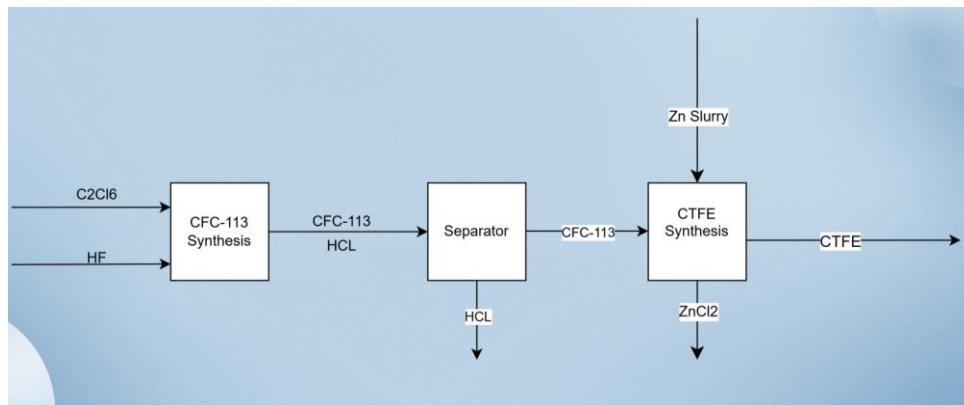


Figure 1: BFD of First Method in Producing CTFE

A second method in producing CTFE involves reacting CFC-113 with a gas mixture of trifluoroethane, ethylene, hydrogen chloride, and oxygen. This method produces no significant waste, which helps to lower our waste treatment cost. However, this method had only been applied in lab-scale experiments, making an industry-scale application unrealistic.

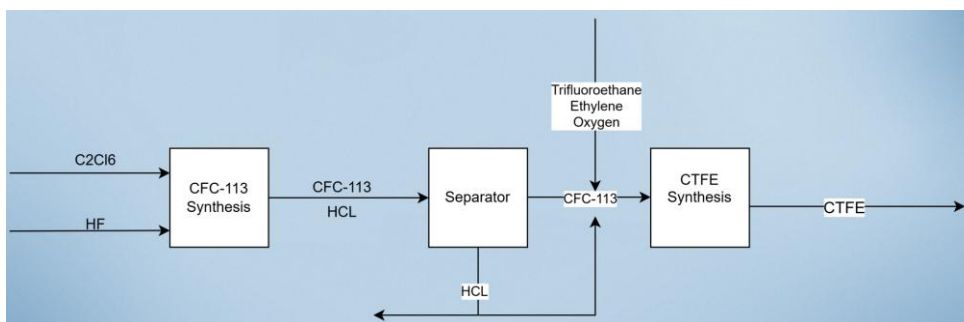


Figure 2: BFD of Second Method in Producing CTFE

The next two processes will focus on the production of methyl chloride. The first method is the hydrogenation of CO_2 , creating water and methanol. The mixture is then reacted with hydrogen chloride to give us methyl chloride. With this method, CO_2 is abundant and cheap, which helps to lower our raw material cost. However, the method produces water as a side-product, which could decay the catalysts in the reactor more quickly.

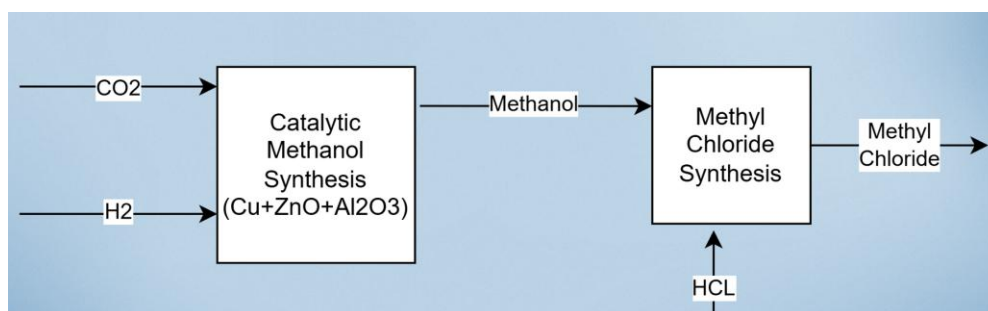


Figure 3: BFD of First Method in Producing Methyl Chloride

The second method instead focuses on the hydrogenation of CO instead of CO_2 . Since there is one less oxygen atom, this reaction does not produce water, which helps to lengthen our catalysts. However, CO for this process must be very pure, which could lead to high raw material cost if we're purchasing from an upstream source, or high operation cost if we decide to purify the CO .

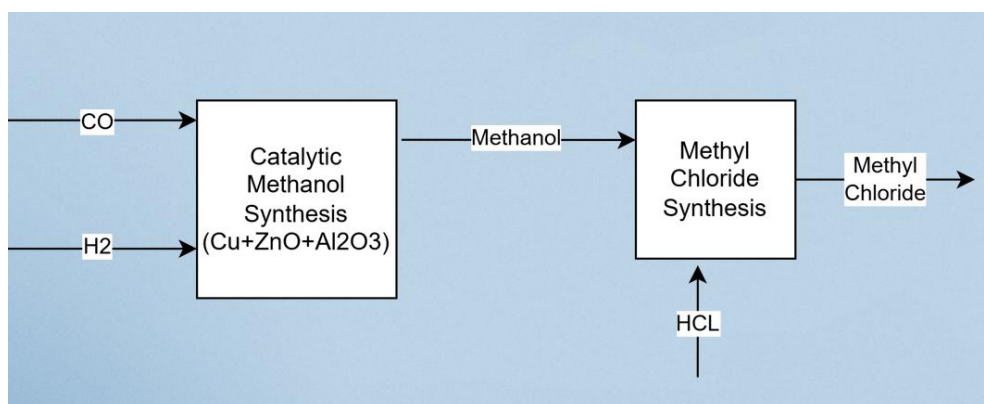


Figure 4: BFD of Second Method in Producing Methyl Chloride

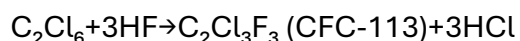
Chosen Process

After the considerations made within the synthesis of our feed stock we have arrived at the following process chain for the production and separation of 1234yf from our desired feed stock

CFC-113

CFC-113 (1,1,2-Trichloro-1,2,2-trifluoroethane) is produced as a key intermediate for the later production of CTFE and eventually 1234yf. This process unit converts hexachloroethane and anhydrous HF into CFC-113.

The reaction that takes place in this step is a highly exothermic halogen exchange where 3 chlorines are replaced on hexachloroethane with fluorine that produces our desired CFC-113 along with stoichiometric amount of anhydrous hydrogen chloride and under fluorinated species.



This stream will be sent to an acid gas flash drum where the unreacted HF and produced HCl are removed from the rest of the fluoro-chloro organic species. The acid that is separated where the HF can be recycled into this process or sent to our second fluorination reaction which we will discuss later. The remaining fluoro-chloro organic species are distilled and the condensed vapor will be high purity CFC-113.

Depending on the desired reaction temperature due to the exothermic nature of the reaction can be performed in either a CSTR or tubular reactor, further understanding of the reaction kinetics is necessary to determine which reactor will function with the highest conversion and efficiency.

CTFE

CTFE (chlorotrifluoroethylene) is one of the two feedstocks that are necessary for the production of 1234yf. CTFE undergoes a dehalogenation reaction in the presence of zinc metal, producing the desired CTFE along with zinc chloride as a byproduct.



In this process zinc metal is not directly mixed with the CFC-113 but instead a slurry is formed with methanol. This slurry has the advantage of allowing for easier processing of the zinc metal and resulting zinc chloride along with greater temperature control of the process as any halogen reaction is typically categorized as exothermic.

The gaseous CTFE will exit through the top of each reactor as gas phase along with other species that may have vaporized in the reactor which will be later separated. A chain of multiple reactors are used in series with the CTFE removed from each and the unreacted zinc slurry and CFC-113 are moved to each successive reactor in order to increase the conversion of the feedstock. In order to select the number of reactors for this conversion a simulation of the process is necessary to determine residence time and sizing.

Now we are able to separate the species and purify the CTFE, this is preformed in two flash drums and a distillation column. The CTFE leaving each reactor will enter the first drum where light flurochlor organics and any acids are separated off leaving the CTFE and heavies to be sent to a distillation column. These heavies are distilled off leaving the CTFE vapor to exit the top of the column and go on to the 1233xf and 1234yf process trains. The heavies are then sent back to the first reactor as they can be converted to the desired CTFE. Additionally, the resulting slurry has the methanol separated and recycled to the slurry prep tank with the zinc chloride being removed as a byproduct.

Methanol

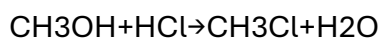
Producing methanol for this process is explored in an effort to offset the environmental impact of the process. This unit take carbon dioxide and hydrogen and is later used in the production of methyl chloride. In this unit, the feed reacts over a $\text{Cu/ZnO/Al}_2\text{O}_3$ catalyst to form methanol and water via a catalytic hydrogenation reaction.



The reaction is carried out in a fixed bed catalytic reactor operating at high pressure and moderate temperature to form methanol. Due to the limitations of the equilibrium during a single pass of the catalyst reactor the unreacted feed is separated in a drum and is sent back to the reactor.

Methyl Chloride

Methyl Chloride is the other of the two feedstocks that are necessary for the production of 1234yf. After the methanol is separated from the water that is produced in the reaction methanol and anhydrous hydrogen chloride are fed into two reactors.

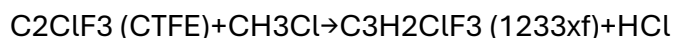


Methanol is directly fed into both of these reactors where as HCl is first fed into one reactor and the unreacted HCl is sent into the second. This is done to minimize the formation of dimethyl ether. The reactor of the effluent consists primarily of the desired methyl chloride with water, methanol, HCl and DME.

This stream is sent through as series of scrubber towers first water, then caustic, finally acid. In the water scrubber methanol and some of the HCl is removed. The caustic scrubber neutralizes and removes the remaining HCL. Finally the acid scrubber dries the now purified MeCl by removing the remaining water content.

1233xf

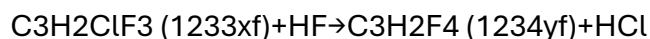
The fluoro-chloro intermediate, 1233xf (2-chloro-3,3,3-trifluoropropene) is produced by reacting the two prepared and purified feedstocks of CTFE and MeCl in a high temperature tubular gas phase reactor. This step converts the C₂ fluorinated intermediate into a C₃ fluoro-chloroolefin which will be later fluorinated to produce 1234yf.



The reactor effluent is quenched once leaving the reactor to avoid producing unwanted side reactions such as cracking or polymerization. This is then sent to be separated from byproducts in a drum where the bottom fluoro-organic product.

1234yf

1234yf is produced by the gas-phase fluorination of the fluoro-chloro intermediate 1233xf using anhydrous hydrogen fluoride. This reaction is the final process in the chain and will directly affect the overall purity and yield. The reaction selectively replaces the remaining chlorine atom on 1233xf with fluorine to form 2,3,3,3-tetrafluoropropene (1234yf), while generating hydrogen chloride as a byproduct.



The reaction is carried out in the gas phase over a chromium based fluorination catalyst. The catalyst consists of chromium oxide that is fluorinated in situ with HF prior to operation to form the active catalytic phase. The reactor is designed as a fixed-bed tubular unit, with catalyst packed inside furnace-heated tubes.

Vaporized 1233xf and excess HF are preheated before entering the reactor to prevent condensation and ensure stable gas-phase operation. HF is used in excess in the process to both drive the fluorination but to keep the catalyst active.

The reactor effluent exits the fluorination reactor as a mixed gas stream containing 1234yf, hydrogen chloride, unreacted HF, and trace amounts of byproducts such as 245fa and light fluorinated hydrocarbons. Immediately downstream of the reactor, the effluent is rapidly quenched to suppress further reaction and limit byproduct formation. The quenched stream is then sent to downstream separation, where acid gases are removed from the organic product stream. Unreacted HF is recovered and recycled back to the reactor inlet to improve overall fluorine utilization, while the remaining organic stream is further processed to isolate high purity 1234yf.

Appendix

Design Basis

The plant location will be in Michigan, US for multiple reasons. The high level of reactivity of R-1234yf give it one of the main advantages of a low GWP, but it is also what makes it dangerous to transport. For this reason, the plant location prioritizes a shorter transport to areas of use over proximity to raw materials. When completing the market analysis, of the approx. 80% of cars that are assembled in North America for the US market, the majority are assembled in the US. Michigan, with port access, and a central location within the states makes a strong location to build the plant. Additionally, while the competitors have southern locations within the US, giving them an advantage on transportation to Mexico, our company can focus on transportation to Canada. As well, being in Michigan means we will have a close proximity to General Motors which is a strong market advantage.

The plant will be at scale to manufacture 1.3M kg/year of R-1234yf. This is to supply approx. 25% of the North American OEM demand. Refer to the material balance to reference the amount of raw materials needed.

Block Flow Diagram

Shown below is a color-coded BFD of our chosen process

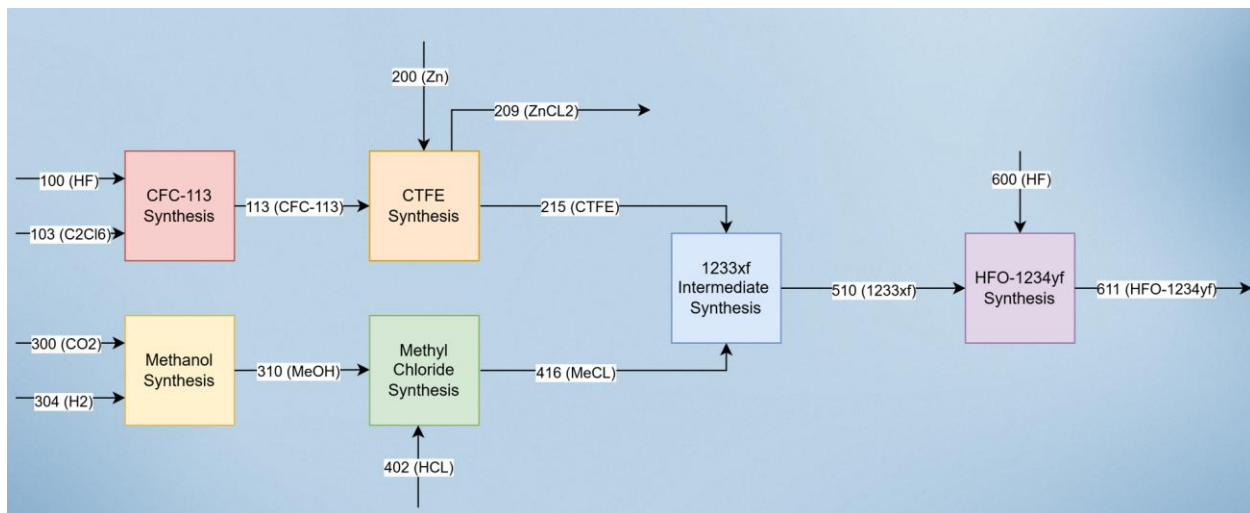


Figure 4: Color-coded BFD of the Main Process

Process Flow Diagram

The process flow diagram will follow the same color-coded system as the BFD. The entire PFD of our process will be lengthy, making it challenging to see what's happening. Therefore, our PFD will be divided by colors. Below is the PFD for CFC-113 synthesis.

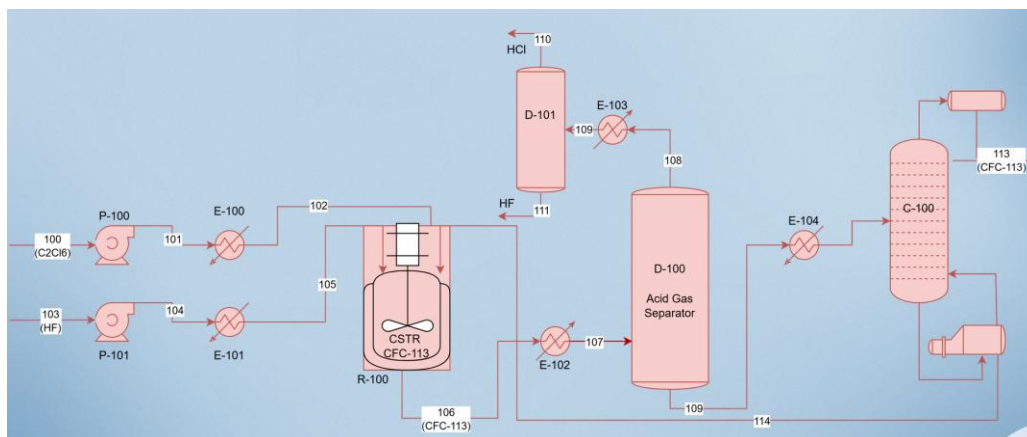


Figure 5: PFD of CFC-113 Synthesis

In Figure 5, C_2Cl_6 and HF first go through a series of pumps and heaters to match the reactor's operating pressure and temperature, respectively. The two species then enter the CSTR reactor and react, forming CFC-113, HCl , and unreacted HF . The product mixture then travels to a separator, where HCl and HF are isolated, and CFC-113 is sent to a distillation column for further purification. Pure CFC-113 can now be used as a reactant for CTFE synthesis, which follows the following PFD:

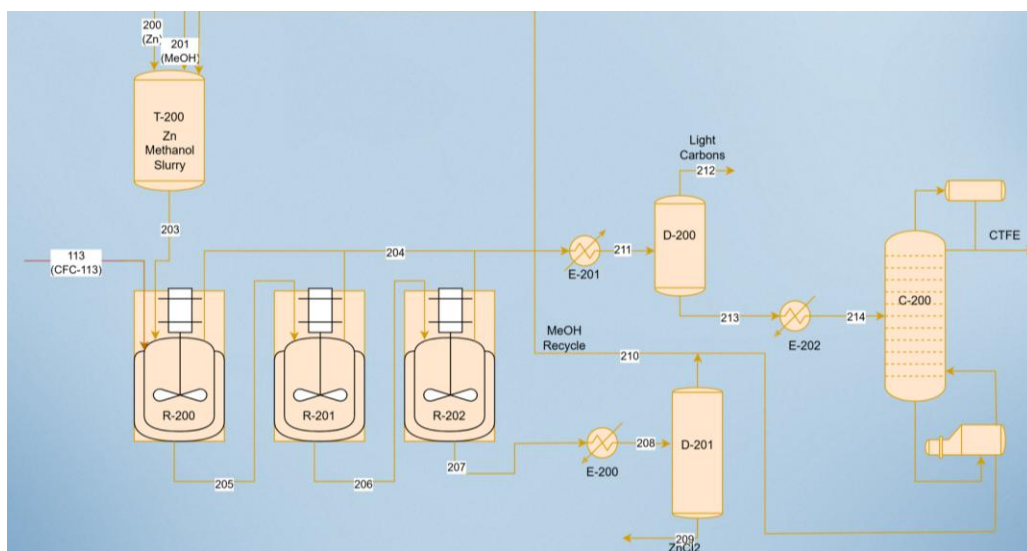


Figure 6: PFD of CTFE Synthesis

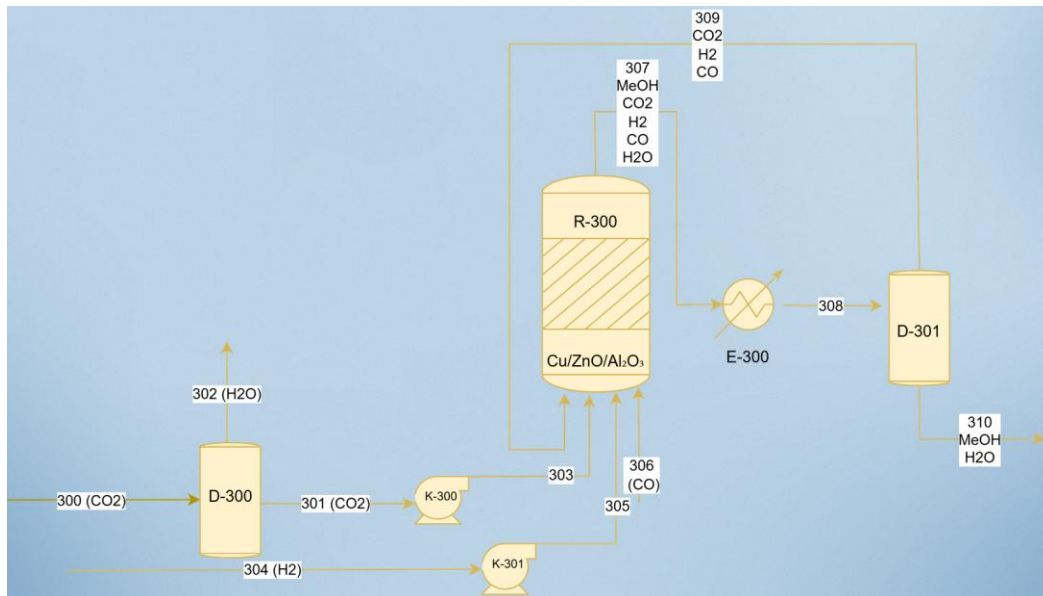


Figure 7: PFD of Methanol Synthesis

In Figure 7, CO_2 was first passed through a knock-out drum to remove any moisture and is then passed through a compressor along with pure and anhydrous H_2 before entering the catalytic reactor R-300. Coming out of the reactor will be a gas mixture of $MeOH$, CO_2 , H_2 , CO , and H_2O , to which the mixture is passed through a condenser to condense the $MeOH$ and H_2O before entering a gas-liquid separator to separate $MeOH$ and H_2O to be used for the next stage, while the gases are recycled back into the catalytic reactor. Figure 8 below describes the next stage.

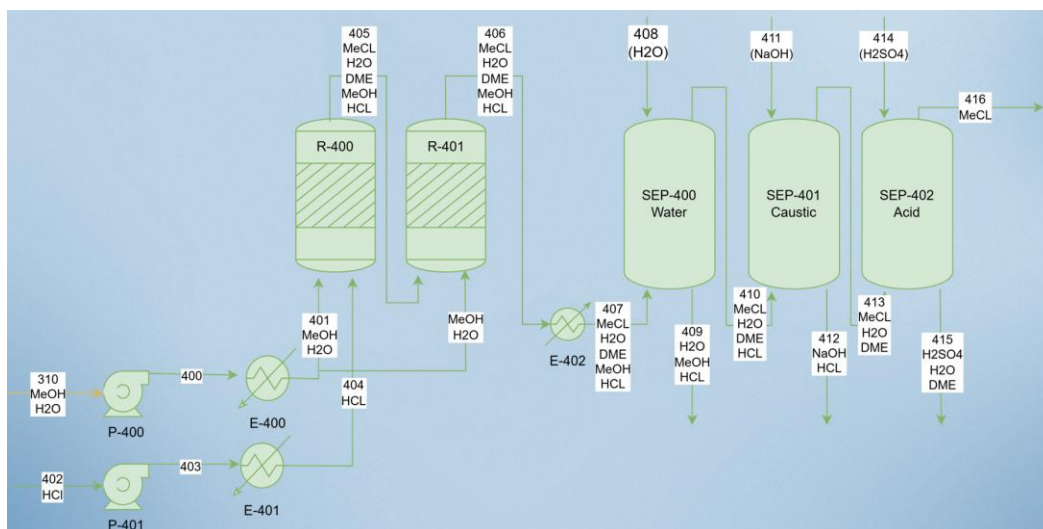


Figure 8: PFD of Methyl Chloride Synthesis

After $MeOH$ and H_2O have been separated, the mixture and anhydrous HCl then go through a series of pumps and heaters before a portion of $MeOH$ and H_2O and all of the HCl is sent to the first reactor, while the second portion is sent to a second reactor. The main purpose of this is to avoid the by-product dimethyl ether (DME). The mixture of $MeCl$, H_2O , DME , $MeOH$, and HCl then passed through a water scrubber tower to remove $MeOH$ and HCl . The mixture then goes through a caustic scrubbing tower to remove remaining HCl . Finally, the mixture goes through an acid scrubbing tower to remove water.

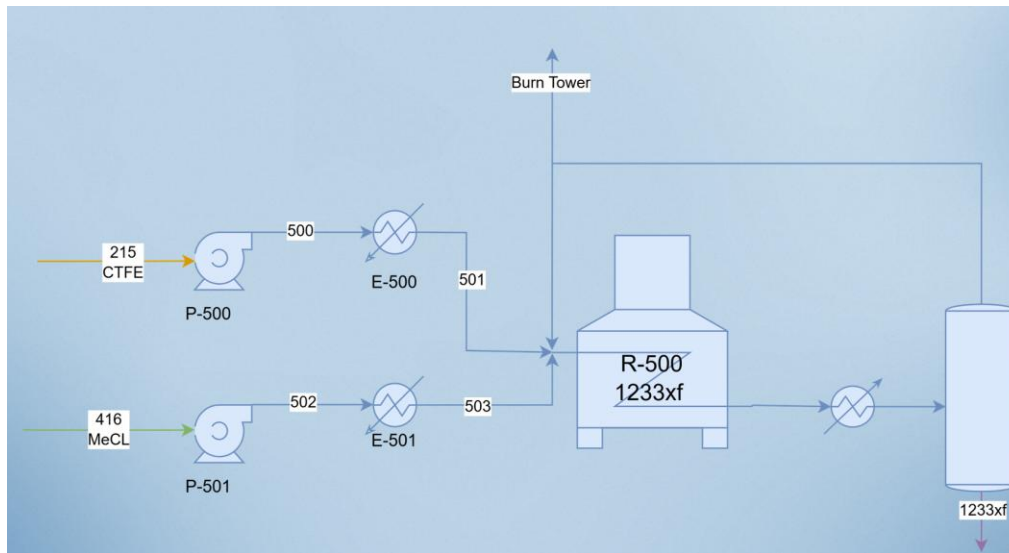


Figure 9: PFD of 1233xf Synthesis

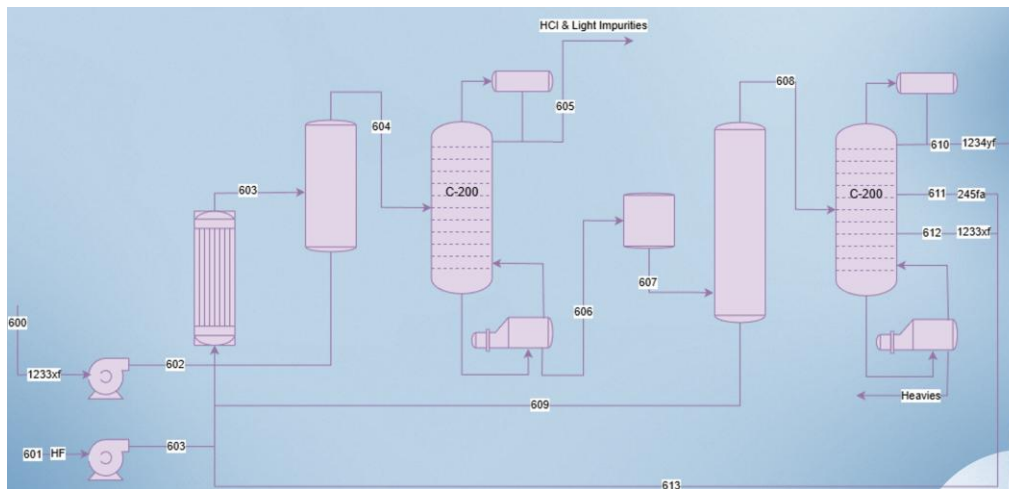


Figure 10: PFD of R-1234yf Synthesis

Material Balance

Although the design basis was in terms of metric tons per year, our material balance will be in terms of kilograms per hour just to make it easier to visualize. Below is the color-coded BFD with material balance.

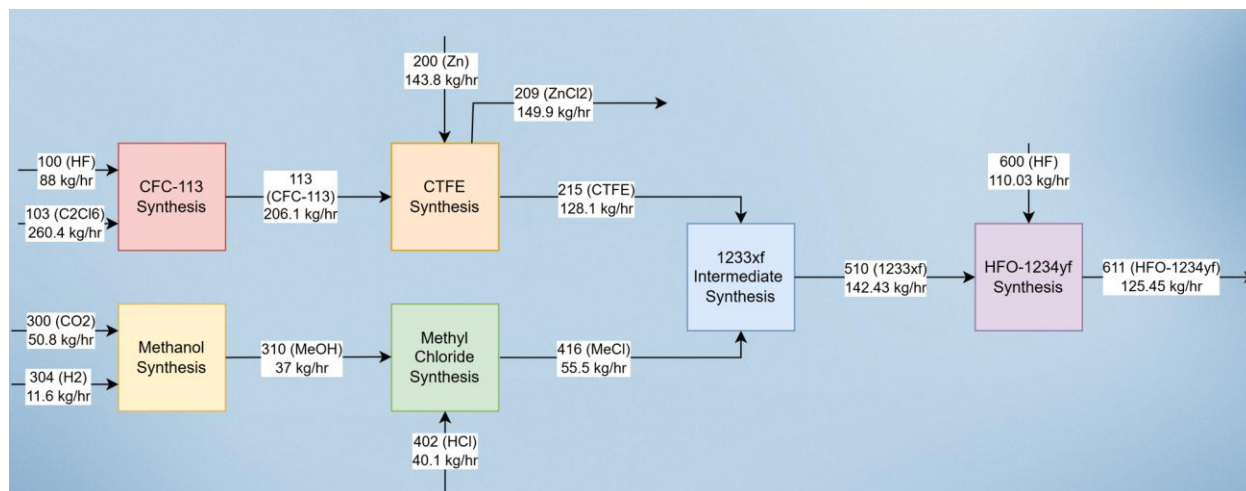


Figure 11: BFD with Material Balance

To perform the material balance, our design basis of 1 million kg R-1234yf per hour is first divided by 8000 operating hours to obtain the amount in kg per hour. Dividing this value by the molar mass of R-1234yf gives our flow rate in kmol per hour, which is important because the chemical reactions are modeled based on mole quantity rather than mass. The reaction for producing R-1234yf is as follows:



Although all species follow a 1:1 relationship, the mole amount of HF is set to be four times of other species to ensure all of the 1233xf had reacted. The calculated mole amounts are then converted back to mass amounts using the corresponding molar mass. Doing this gets us the material balance for the “HFO-1234yf” block in Figure 11. Next, the synthesis of 1233xf follows the reaction:



The species in this reaction also follow a 1:1 relationship, but since we’re producing both CTFE and MeCl, their mole amounts also follow a 1:1 relationship. Using each species’ corresponding molar mass, we can convert the mole amount back to mass amount, giving us the material balance for “1233xf Intermediate Synthesis”.

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